Q5	Identify open ports/services with nmap and disable the unnecessary ones	5 Marks

Objective: Discover which ports are actually open, then reduce the attack surface by disabling services you do not need.

Step-by-Step Algorithm

Step	Phase	Action	Command / Tool	What You Should See
1	TEST	Scan the host for open TCP ports.	`nmap -sT -p- 127.0.0.1`	Open-port list
2	ANALYSE	Match each open port to a running service.	ss -tuln	Port-to-service map
3	ANALYSE	Decide which services are unnecessary.	—	Disable list
4	CONFIGURE	Stop and disable each unneeded service.	`sudo systemctl disable --now <svc>`	Service inactive
5	TEST	Re-scan to confirm the port is now closed.	`nmap -sT -p- 127.0.0.1`	Fewer open ports

Phase key: ● SETUP · ● CONFIGURE · ● EXECUTE · ● TEST · ● ANALYSE · ● REPORT · ● FINISH

Worked Example (with sample values & output)

Applying the steps to lab-vm

The first scan finds an unused Apache service on port 80. After disabling it, the re-scan shows only SSH remaining open, shrinking the attack surface.

Port	Service	Needed?	Action taken
22/tcp	SSH	**Yes**	Keep
80/tcp	Apache (unused)	**No**	Disable service

Sample output:

```
Before:  22/tcp open   80/tcp open
After :  22/tcp open
```

Reusable Template

Open-ports record template

Port	Service	Needed? (Y/N)	Action
22/tcp	SSH	Yes	keep
80/tcp	Apache(unused)	no	Disable service(sudo systemctl disable --now apache2)

Your Output / Screenshot (paste result here):

```
┌──(kali㉿kali)-[~]
└─$ nmap -sT -p- 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-11-03 02:51 -0500
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00016s latency).
All 65535 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 65535 closed tcp ports (conn-refused)

Nmap done: 1 IP address (1 host up) scanned in 2.57 seconds

┌──(kali㉿kali)-[~]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ ss -tuln
Netid     State      Recv-Q     Send-Q         Local Address:Port         Peer Address:Port
```

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl disable --now apache2

┌──(kali㉿kali)-[~]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ systemctl status apache2
○ apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
     Active: inactive (dead)
       Docs: https://httpd.apache.org/docs/2.4/

┌──(kali㉿kali)-[~]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ nmap -sT -p- 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-11-03 03:05 -0500
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00020s latency).
All 65535 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 65535 closed tcp ports (conn-refused)

Nmap done: 1 IP address (1 host up) scanned in 3.64 seconds
```

Before: 22/tcp open 80/tcp open

After : 22/tcp open

Port	Service	Needed?	Action taken
22/tcp	SSH	Yes	Keep
80/tcp	Apache (unused)	No	Disable service

Scanning with **nmap** revealed the open ports, and **ss -tuln** mapped each one to a running service. The unneeded Apache service on port 80 was stopped and disabled with **systemctl** disable --now. The re-scan showed only SSH (22/tcp) open, so the attack surface was reduced.